



Analyzing Prioritized Environmental Issues in Port Operations Using the q-ROF-SWARA Method

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Abstract: This study addresses the impact of increasing environmental pressures on a global scale on the long-term strategies of businesses, with particular emphasis on the importance of environmental responsibility in port operations. Port activities are directly associated with various environmental issues, including climate change, global warming, air and water pollution, noise pollution, waste management, and energy consumption. Effectively identifying and prioritizing these issues is critical not only for protecting environmental and social well-being but also for enhancing the operational performance and competitiveness of port enterprises. In this context, the aim of the study is to evaluate the priority environmental issues faced by port operations using an analytical approach. To this end, the q-rung orthopair fuzzy step-wise weight assessment ratio analysis (q-ROF-SWARA) method, one of the multi-criteria decision-making techniques, was employed due to its ability to effectively handle uncertainty and subjective expert judgments. The findings indicate that energy consumption is the most significant environmental issue for port operations, while noise is considered the least important relative to other factors. The results provide valuable insights for decision-makers in developing sustainable port management practices and formulating effective environmental strategies.

Keywords: Port operations; Priority environmental issues; Multi-criteria decision making; q-rung orthopair fuzzy step-wise weight assessment ratio analysis method

1 Introduction

Carrying out sustainable development and environmental protection activities is one of the greatest challenges faced by societies and economies today. Throughout history, humanity has perceived natural resources as if they were unlimited, leading environmental problems to assume a global dimension. Increasing industrialization, population growth, and urbanization have further intensified pressure on natural ecosystems, thereby accelerating environmental degradation. In this context, it has become evident that environmental problems have a multidimensional structure that requires attention not only at the national level but also on a global scale. Due to the international nature of these problems, it has become necessary for countries to develop joint policies and establish corresponding implementation plans [1]. In maritime transport, the concept of sustainability regulates maritime activities in a way that minimizes harm to the marine ecosystem by reducing negative environmental impacts, such as emissions across the supply chain, while maintaining economic interests. Reducing the environmental impacts of ports and promoting a lasting ecological balance are also among the objectives of this concept [2]. In particular, in order to prevent the degradation of natural resources and ensure environmental sustainability, it is of great importance for ports to continuously improve their quality standards in order to maintain their presence in an increasingly competitive global environment. In this context, the adoption of environmentally friendly technologies, the implementation of infrastructure investments that enhance energy efficiency, and the widespread application of emission reduction practices are essential. Moreover, by strengthening their technical capacities, ports can provide cleaner, safer, and more efficient services, while integrating innovative solutions such as digitalization, automation, and smart port applications into their operations [3]. On the other hand, establishing a strong connection with their hinterland, maintaining relationships with nearby ports, and integrating available resources into the port system to create a highly efficient port network are critical factors for

ensuring the sustainability of ports [4]. However, the rapid development of ports also brings certain environmental problems; air, water, and noise pollution are becoming increasingly widespread, and environmental concerns are growing. In particular, increasing vessel traffic, cargo handling activities, and port-related industrial operations place pressure on ecosystems, threatening the sustainability of marine and coastal environments [5]. Today, businesses are under social pressure to adopt an environmentally oriented approach. These pressures have led to increasing concern that businesses must take sustainability and environmental factors into account in their production models and business operations [6]. Accordingly, port authorities aim to find a balance between economic, social, and environmental dimensions in order to ensure the sustainable development of the local community surrounding the port [7]. In this context, sustainability in ports is considered as establishing a balance between business systems and practices that protect human health and environmental values while ensuring the continuity of economic activities. This approach encompasses not only the reduction of environmental impacts but also the fulfillment of social responsibilities and the long-term sustainability of economic performance. Therefore, adopting a holistic sustainability perspective that takes into account the expectations of all stakeholders plays a critical role in creating lasting value within the port sector [8]. There are three main sources of environmental risks in ports: risks arising from port operations, risks originating from ships, and risks caused by off-port vehicles in intermodal transportation networks. Some of the problems caused by these sources include air pollution, habitat degradation and loss, harm to biological species, pollution from various waste sources (oils, wastewater and stormwater discharges, hazardous material waste, bilge water, ballast water, waste from ship paint maintenance, solid waste, and waste from port infrastructure and superstructure maintenance, etc.), traffic congestion, noise and light pollution, loss of cultural resources, soil and water contamination, oil spills, and soil erosion [9].

From this perspective, the major environmental issues arising in ports should not be viewed merely as factors affecting operational activities, but rather as multidimensional and dynamic parameters with wide-ranging impacts extending from ecological balance to social welfare. These issues are directly related to environmental components such as air and water pollution, noise, waste management, energy consumption, degradation of marine ecosystems, and carbon emissions. Therefore, managing these environmental impacts within the framework of sustainability is of great importance for port authorities not only in terms of economic efficiency but also with respect to social responsibility and the principles of environmental justice. Consequently, the systematic identification, analysis, and prioritization of key environmental problems arising in ports are of critical importance for building a sustainable society. Identifying these environmental issues emerges as a strategic requirement for transforming business processes into a sustainable structure. Minimizing the major environmental problems in ports helps optimize resource use while simultaneously generating economic and social benefits for society.

In this context, the development of indicators for measuring and monitoring environmental performance supports more effective and transparent decision-making processes. Moreover, strong collaboration among public authorities, the private sector, and local communities is essential for port managements to achieve their sustainability goals. The integration of innovative technologies and environmentally friendly practices into port operations is considered an important tool for reducing environmental risks. Finally, increasing environmental awareness and embedding a sustainability culture at the institutional level contribute significantly to improving the long-term environmental and social performance of ports. In line with all these considerations, this study aims to systematically identify the major environmental issues encountered in port operations operating within the province of Samsun. Within this framework, the environmental impacts arising from port activities are examined in detail, and the problem areas that are critical in terms of sustainability are identified. The environmental issues determined in the scope of the research are analyzed using multi-criteria decision-making (MCDM) methods and ranked according to their levels of impact. In this way, the study seeks to contribute to enabling decision-makers to develop environmental management strategies in a more rational and effective manner. In addition, the study strengthens its theoretical foundation by providing a comprehensive review of the main environmental issues arising in ports, the concepts related to these issues, and the methods proposed in the existing literature. Through the literature review, the scientific basis of the study is reinforced and the validity of the applied methodologies is supported. In the final section, the findings obtained from the analyses are presented in detail, and the potential contributions of these findings to port operations and relevant stakeholders are discussed. Finally, based on the results obtained, recommendations for industry practices are developed, and evaluations that may serve as guidance for future academic research are provided.

2 Literature

Below are some studies in the national and international literature addressing the major environmental issues emerging in ports.

Wiegman and Geerlings [10] examined sustainable port innovations and the barriers and facilitating factors affecting the successful implementation of port practices; Battistelli et al. [11] analyzed existing regulations in port areas and proposed sustainable actions to address the environmental impacts of maritime activities; del Saz-Salazar et al. [12] emphasized the necessity of measuring the adverse effects resulting from port growth; Pozo [13] investigated

the recurring atmospheric emission problems caused by maritime activities at the Port of Santos; Murphy and King [14] studied the noise exposure levels of residents living around Dublin Port in Ireland; Gómez et al. [15] proposed a standardized and integrated procedure to assess environmental risks at the level of pollutant sources in port water systems; Sislian et al. [16] conducted a literature review on port sustainability and the ocean carrier network issue; Venturini et al. [17] examined the Berth Allocation Problem, an optimization issue that assigns berthing times and locations to ships at container terminals; Abbasi and Pishvae [18] proposed a two-stage optimization model for the dry port location problem in Iran; Lagoudaki et al. [19] conducted an environmental risk assessment for ports and provided various recommendations; Çalışkan [20] compiled potential challenges in smart port transformation and developed a hierarchical model to analyze the relationships among these challenges; Čurović et al. [21] investigated the impact of COVID-19 on environmental noise generated from ports; Tok and Bucak [22] examined the effects of port congestion on sustainability and evaluated its potential impacts on port operations and the supply chain; Zeng et al. [23] analyzed carbon emission reduction within the scope of integrated logistics at the Port of Shanghai; Ayaz and Bucak [24] investigated the causes and most frequent periods of port congestion, as well as measures to mitigate congestion and strategies that could be developed and Yuan et al. [25] conducted a bibliometric analysis focusing on port emission reduction.

A detailed review of the literature reveals that academic research on the challenges and environmental issues faced in ports remains quite limited. Most existing studies are confined to specific sectors, port types, or countries, and holistic approaches that address multiple dimensions simultaneously are insufficiently developed. In particular, comprehensive analyses that thoroughly examine the underlying causes of barriers, their interrelationships, and their operational, environmental, and social impacts are scarce. This situation restricts the accumulation of both theoretical and practical knowledge and results in a lack of studies that can serve as effective guidance for decision-makers. In this context, conducting a comprehensive study aimed at addressing this gap can make a significant contribution to the literature. Such research would not only help reduce existing knowledge deficiencies but also enable the systematic identification, classification, and prioritization of key barriers encountered in ports. Moreover, the findings are expected to support the development of more effective and sustainable solutions for port managers and policymakers. From this perspective, the study has the potential to serve as a valuable reference for both academic research and industry practices.

3 Methodology

It is common for decision-making problems to involve conflicting criteria and uncertainties. In such problems, there are differences in the priority or importance levels of the criteria considered. Weighting techniques are used in the process of determining the importance levels of criteria. Among the methods employed for this purpose, step-wise weight assessment ratio analysis (SWARA) features a straightforward, easily understandable and applicable structure. In this study, an extension of SWARA, defined under q-rung orthopair fuzzy sets (q-ROFSs), enables the determination of criteria weighting coefficients using uncertain information. In the following section, information about q-ROFSs will be presented first. Then, the q-ROF-SWARA process steps will be presented.

Let X be a discourse universe. A q-ROFSA of X is provided in Eq. (1), where $a_A(x)$ denotes the membership degree and $b(x)$ depicts the non-membership degree.

$$A = \{ \langle x, (a_A(x), b_A(x)) \rangle \mid x \in X \} \quad (1)$$

There are the following conditions regarding membership level and non-membership level in q-ROFSs: $a_A(x) \in [0, 1], b_A(x) \in [0, 1], 0 \leq (a_A(x)^q + b_A(x)^q) \leq 1, (q \geq 1)$. Also, the uncertainty membership degree can be written as $\pi_A(x) = (a_A(x)^q + b_A(x)^q - a_A(x)^q b_A(x)^q)^{1/q}$.

For convenience, let $g_1 = \langle a_1, b_1 \rangle$ and $g_2 = \langle a_2, b_2 \rangle$ be two q-ROF numbers (q-ROFNs). Eqs. (2)–(6) displays the fundamental operations, Eq. (7) represents the score function ($\mathcal{S}(g_1)$), and Eq. (8) gives the accuracy function ($\mathcal{A}(g_1)$) [26–28].

$$g_1 \oplus g_2 = \left(\sqrt[q]{(a_1)^q + (a_2)^q - (a_1)^q (a_2)^q}, b_1 b_2 \right) \quad (2)$$

$$g_1 \otimes g_2 = \left(a_1 a_2, \sqrt[q]{(b_1)^q + (b_2)^q - (b_1)^q (b_2)^q} \right) \quad (3)$$

$$\xi_{g_1} = \left(\sqrt[q]{1 - (1 - (a_1)^q)^\xi}, (b_1)^\xi \right), \xi \geq 0 \quad (4)$$

$$g_1^\xi = \left((a_1)^\xi, \sqrt[q]{1 - (1 - (b_1)^q)^\xi} \right), \xi \geq 0 \quad (5)$$

$$(g_1)^c = (b_1, a_1) \quad (6)$$

$$S(g_1) = \frac{1}{2} ((a_1)^q - (b_1)^q + 1), S(g_1) \in [0, 1] \quad (7)$$

$$\mathcal{A}(g_1) = (a_1)^q + (b_1)^q, \mathcal{A}(g_1) \in [0, 1] \quad (8)$$

Let $g_j = \langle a_j, b_j \rangle$ be a collection of q-ROFNs, where $j = 1, \dots, n$. Eq. (9) gives the q-ROF weighted arithmetic average (q-ROFWAA) operator and Eq. (10) provides the q-ROF weighted geometric average (q-ROFWGA) operator [26–28].

$$\text{q-ROFWAA}(g_j) = \left\langle \left(1 - \prod_{j=1}^n (1 - a_j^q)^{w_j} \right), \prod_{j=1}^n b_j^{w_j} \right\rangle \quad (9)$$

$$\text{q-ROFWGA}(g_j) = \left\langle \prod_{j=1}^n a_j^{w_j}, \left(1 - \prod_{j=1}^n (1 - b_j^q)^{w_j} \right) \right\rangle \quad (10)$$

where, $W = (w_1, \dots, w_n)^T$ is the weight vector, $0 \leq w_j \leq 1$, and $\sum_{j=1}^n w_j = 1$. The following steps gives details of q-ROF-SWARA [29–31].

Step 1. The sets of criteria and experts are determined. $C = \{C_1, \dots, C_j, \dots, C_n\}$ denotes the set of criteria, while $E = \{E_1, \dots, E_k, \dots, E_z\}$ depicts the set of experts.

Step 2. The weight coefficients of experts λ_k are determined, where $k = 1, \dots, z$, $0 \leq \lambda_k \leq 1$ and $\sum_{k=1}^z \lambda_k = 1$.

Step 3. Experts evaluate the importance of criteria using linguistic expressions provided in Table 1.

Table 1. The linguistic expressions and their corresponding q-ROF numbers (q-ROFNs) equivalents

Linguistic Expressions	Notations	q-ROFNs	
		a	b
Extremely High	EH	0.99	0.20
Very Very High	VVH	0.90	0.25
Very High	VH	0.80	0.30
High	H	0.70	0.40
Medium High	MH	0.60	0.45
Medium	M	0.50	0.50
Medium Low	ML	0.45	0.60
Low	L	0.40	0.70
Very Low	VL	0.30	0.80
Very Very Low	VVL	0.25	0.90
Extremely Low	EL	0.20	0.99

As a result, $\iota_j^{(k)} = (a_j^{(k)}, b_j^{(k)})$ denotes the importance ratings of k -th expert for j -th criterion.

Step 4. Eq. (9) is employed to produce the q-ROF integrated importance of j -th criterion ι_j .

Step 5. Eq. (7) is used to determine the crisp values of ι_j .

Step 6. The criteria are ranked in descending order according to their $S(\iota_i)$ values. After that, the comparative importance score of each criterion is determined. In this context, $c_1 = 1$ is assigned to the most important criterion.

Step 7. The comparative importance of each criterion's score value (c_j) is calculated using the importance ranks of the criteria. c_1 represents the comparative importance score of the most important criterion, and c_n represents the comparative importance score of the least important criterion. The c_2 value is calculated by subtracting the $S(\iota_j)$ value of the most important criterion from the $S(\iota_j)$ value of the second most important criterion. A similar procedure is followed for the other criteria. On the other hand, $c_1 = 1$ is allocated to the most crucial criterion.

Step 8. Eq. (11) produces the comparative coefficient (h_j) of each criterion:

$$h_j = \begin{cases} 1, & \text{if } j = 1, \\ c_j + 1, & \text{if } j > 1. \end{cases} \quad (11)$$

Step 9. Eq. (12) gives the recalculated coefficients (q_j).

$$q_j = \begin{cases} 1, & \text{if } j = 1, \\ \frac{h_{(j-1)}}{h_j}, & \text{if } j > 1. \end{cases} \quad (12)$$

Step 10. Eq. (13) is employed to calculate weight coefficients of criteria.

$$w_j = \frac{q_j}{\sum_{j=1}^n q_j} \quad (13)$$

where, $0 \leq w_j \leq 1$ and $\sum_{j=1}^n w_j = 1$.

4 Results

In this study, the opinions of six operations managers were consulted in order to determine the relative importance levels of the environmental criteria related to port operations. Based on their professional experience and expertise, the managers evaluated the criteria and assigned importance scores to each criterion within the decision-making process. The evaluations obtained were used as inputs in the analysis and contributed to establishing the importance ranking of the criteria. Detailed information regarding these evaluations is presented in Table 2.

Table 2. The linguistic importance evaluations of criteria provided by experts

Criteria	Source	Criteria
Climate change	[24], [32]	C1
Air and water quality	[16]	C2
Energy consumption	[1]	C3
Noise	[1], [15]	C4
Ship-generated waste	[32]	C5
Relations with local communities	[32]	C6
Port development related to land and water	[32]	C7
Solid waste/Port waste	[1]	C8

In this study, experts evaluated the importance levels of the criteria using the linguistic expressions presented in Table 1. These linguistic terms allowed expert opinions to be expressed in a more flexible manner that reflects uncertainty. The qualitative evaluations provided by the experts were subsequently converted into numerical values for use in the analysis process. The final evaluation results of the criteria are presented in detail in Table 3.

Table 3. The linguistic importance of criteria

Experts	Criteria							
	C1	C2	C3	C4	C5	C6	C7	C8
E1	M	M	H	M	M	H	H	M
E2	H	H	VH	L	M	M	M	M
E3	M	M	H	M	M	M	M	H
E4	M	M	H	M	M	M	H	M
E5	H	M	H	M	H	M	M	H

Note: H = High; M = Medium; L = Low.

The calculations regarding q-ROF-SWARA was carried out, and the results displayed in Table 4 were obtained.

According to the results of the analysis, “C3–Energy consumption” was identified as the most important criterion among all the evaluated criteria. This finding indicates that energy usage plays a critical role in port operations in terms of both environmental impacts and economic costs. High levels of energy consumption contribute to increased carbon emissions and accelerated depletion of natural resources, while also directly affecting the operational efficiency of port enterprises. Therefore, developing strategies aimed at improving energy efficiency and integrating renewable energy sources into port activities should be considered a top priority for sustainable port management [32].

Table 4. The weighting results of q-rung orthopair fuzzy step-wise weight assessment ratio analysis (q-ROF-SWARA)

Criteria	ι_j		$S(\iota_j)$	c_j	h_j	q_j	w_j	Rank
	a	b						
C3	0.724	0.377	0.663	0.000	1.000	1.000	0.181	1
C1	0.603	0.457	0.562	0.101	1.101	0.907	0.165	2
C7	0.603	0.457	0.562	0.000	1.000	0.907	0.165	2
C8	0.603	0.457	0.562	0.000	1.000	0.907	0.165	2
C2	0.558	0.478	0.532	0.029	1.029	0.881	0.160	5
C5	0.558	0.478	0.532	0.000	1.000	0.881	0.160	5
C6	0.558	0.478	0.532	0.000	1.000	0.881	0.160	5
C4	0.483	0.534	0.480	0.052	1.052	0.837	0.152	8

Note: ι_j = Integrated q-rung orthopair fuzzy evaluation value; a = Membership degree of q-ROFN; b = Non-membership degree of q-ROFN; $S(\iota_j)$ = Score function value of integrated q-ROF evaluation; c_j = Comparative importance score; h_j = Comparative coefficient; q_j = Recalculated coefficient; w_j = Final weight coefficient.

5 Conclusions

In a globalized world, maritime transport has become one of the fundamental sectors ensuring the continuity of international trade and playing a decisive role in the development of the global economy. However, the rapid growth of the sector has created significant environmental pressures, particularly in coastal and port areas, due to ship-generated waste, greenhouse gas emissions, and various pollutants. This situation has transformed the role of ports from being limited solely to vessel and cargo handling into a multidimensional structure that also encompasses environmental management, sustainability, and energy efficiency. In this context, identifying the major environmental issues arising in ports and developing sustainable strategies to address them are of great importance for both environmental protection and the enhancement of ports' long-term competitiveness.

Accordingly, the present study provides a comprehensive analysis of the main environmental issues encountered in ports operating in the province of Samsun. The results of the analysis indicate that energy consumption is the most critical environmental factor, which is attributed to the high energy demand of port operations, the limited implementation of energy efficiency practices, and the insufficient use of renewable energy sources. Conversely, although noise was identified as the lowest priority environmental issue, this finding does not suggest that its impacts can be entirely neglected in terms of occupational health and environmental sustainability; rather, it highlights the need to address noise within a holistic environmental management framework.

Overall, this study can be regarded as a guiding research effort that addresses a significant gap in the literature by focusing on the major environmental issues encountered in the specified ports. The findings enable port operators to assess environmental impacts in a more systematic manner and support sustainability-oriented decision-making processes. In this respect, the study provides a solid scientific basis for the development of environmental sustainability policies for ports. Moreover, the results offer decision-makers the opportunity to more accurately determine the relative importance of environmental issues and to allocate resources more effectively in line with these priorities. The systematic prioritization of environmental impacts enables port operators to improve their environmental performance while also facilitating the achievement of long-term economic and social benefits. Through this approach, limited resources can be allocated more efficiently to the most critical environmental issues, leading to a more effective and efficient environmental management process. In particular, improvements in areas such as energy consumption, emissions, and waste management contribute not only to the reduction of environmental risks but also to the lowering of operational costs. In this way, a balanced and integrated relationship between environmental sustainability and economic performance can be established. In this context, the findings of the study serve as a strategic reference for port authorities, policymakers, and other relevant stakeholders. The results provide decision-makers with a scientifically grounded roadmap for identifying environmental priorities and developing sustainability-oriented policies. Moreover, integrating these findings into industry practices can help ports fulfill their environmental responsibilities while simultaneously enhancing their competitive capacity. From this perspective, the study represents an important guide for promoting the broader adoption of sustainable port management practices.

In future research, the topic addressed in this study can be examined in a more comprehensive and in depth manner by employing different MCDM methods. In particular, applying multiple methods to the same problem would allow for comparisons among analytical approaches and enable the assessment of the consistency and reliability of the results obtained. Such comparative analyses would contribute to a clearer identification of the strengths and limitations of each method. In addition, existing analyses may be extended and re-evaluated within the framework

of alternative fuzzy logic-based approaches or hybrid multi criteria models. The use of fuzzy methods offers the opportunity to more realistically reflect the uncertainty inherent in decision-makers' evaluations, while hybrid models can produce more robust outcomes by combining the advantages of different techniques. Furthermore, future studies could conduct comparative analyses of ports located in different geographical regions, thereby enhancing the generalizability of the findings. In this regard, such research is expected to provide a more holistic and comparative perspective on the evaluation of ports' environmental performance and to make significant and lasting contributions to the literature on sustainable port management.

Author Contributions

Conceptualization, S.K. and A.G.; methodology, A.A.; software, A.A.; validation, A.A.; formal analysis, A.A.; investigation, S.K. and A.G.; resources, S.K. and A.G.; data curation, A.A.; writing—original draft preparation, S.K., A.A., and A.G.; writing—review and editing, S.K., A.A., and A.G.; visualization, S.K.; supervision, S.K.; project administration, S.K. and A.A.; funding acquisition, S.K. and A.A. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflict of interest.

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